Venkata Lakshmi Parimala Pasupuleti

Senior Analytics Engineer · AI & Data Engineering · LLM Integration · RAG · Context Engineering · MCP · dbt · Python · SQL · AWS · Semantic Layer Design

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | AgentProbe (open-source) | Virginia · Open to Boston, MA

PROFESSIONAL SUMMARY

I build **AI-ready data foundations — semantic layers, RAG pipelines, context engineering for LLMs** — with 9+ years of analytics and data engineering across financial and regulated domains. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Currently at Intuit: owning **text-to-SQL LLM features**, **RAG grounding** on customer financial records, **dbt semantic layer definitions**, and **MCP integrations** for natural-language financial queries; **Dagster-based pipeline orchestration** for LLM workflow automation. Two open-source AI systems: **AgentProbe** — an LLM accuracy measurement framework (5-metric eval engine, 1,240-case CI harness, LLM-as-judge) that directly maps to the JD's "measure their accuracy"; **Leasel** — **semantic data model + context engineering for AI agents** with structured knowledge graph, citation tracing, 8-agent LangGraph StateGraph, and 93.6% precision. At Entrata: 7 years of **data architecture, ETL pipelines, SQL optimization** (200+ queries, 30% perf gain), and **data governance / compliance** for sensitive regulated data (PHI, HUD, HIPAA). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Semantic Layer Design | Context Engineering for LLMs | Text-to-SQL (LLM) | RAG Pipelines | MCP Integrations | LLM Accuracy Measurement | Data Architecture | dbt Data Modeling | ETL Pipelines | Data Governance & Compliance | Analytics Engineering | Cloud Lakehouse (AWS S3)

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — LLM Integration · Text-to-SQL · RAG · Context Engineering · dbt · Dagster · Python · AWS

- Build **text-to-SQL LLM features** integrating Anthropic Claude for natural-language financial queries: users ask questions in plain English, Claude translates intent to structured SQL, and retrieved results are grounded with full citation traceability back to source financial records.
- Design **dbt semantic layer definitions** for financial data pipelines: standardize metric definitions, entity relationships, and metadata contracts so LLM-powered queries retrieve consistent, trustworthy data from a single semantic source of truth.
- Implement **Dagster-based pipeline orchestration** for LLM workflow automation: schedule, monitor, and retry multi-step data + AI pipelines that ingest financial records, apply transformations, and feed downstream LLM inference steps.
- Own **MCP integrations** connecting Claude to financial data APIs; build RAG grounding pipelines on customer financial records; design **context engineering** schemas (structured outputs, metadata tagging) so Claude responses are traceable and validated before surfacing to users.
- Run **LLM accuracy measurement** CI (LLM-as-judge + golden datasets): caught 3 prompt regressions pre-release, zero hallucination incidents post-deploy; containerize on Docker, ship via GitHub Actions to **AWS**, with OpenTelemetry + Grafana for production AI observability.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Data Architecture · ETL Pipelines · SQL · Python · AWS · Data Governance

- Designed **data architecture** and ETL pipelines for a multi-tenant SaaS platform: schema design, data access layers, and query optimization across a PostgreSQL estate serving 100K+ monthly active users and 1M+ property records.
- Optimized 200+ SQL queries using execution plan analysis and indexing — 30% performance improvement; built Python/Bash **automation pipelines** eliminating 80% of manual operational work; RabbitMQ async processing with idempotency for bulk tenant workflows.
- Owned HUD income-limit **data governance** ETL: federal source → 180K regulatory domain records with validation, rollback checks, and compliance enforcement; 7 annual cycles, zero violations — sensitive institutional data at production scale.
- **AWS S3** PHI migration with Libsodium encryption (100% PHI compliance, zero security incidents); **data privacy** and compliance for sensitive resident records; Docker microservices redesign cutting execution time 50%; reviewed 300+ PRs; mentored junior engineers.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management **data pipelines** processing 1.6M HUD/Tax Credit records; led AWS S3 cloud migration from legacy server storage; designed reconciliation workflows with compliance validation and rollback logic for sensitive affordable-housing data.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Analytics Engineering Research · Statistical Modeling · Python · R · HuggingFace · PyTorch

- Built predictive models ($R^2 = 0.977$), applied Isolation Forest anomaly detection across multi-source datasets; fine-tuned classification model (HuggingFace + PyTorch) with full training loop (data prep → fine-tuning → eval → deploy on Quart), presented to GMU faculty panel. Mentored 600+ students on data analytics and statistical modeling.

PORTFOLIO PROJECTS

AgentProbe — LLM Accuracy Measurement & Evaluation Framework

LLM Accuracy · Eval CI · MCP-Ready

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Pinecone, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- Built a **5-metric LLM accuracy measurement framework** (factual accuracy, hallucination rate, retrieval quality, completeness, compliance) with LLM-as-judge scoring — directly answers the JD requirement to "measure their accuracy" for AI agents querying domain data.
- CI-integrated regression detection blocks deploys when accuracy drops; routes prompts across 5 LLM providers via LLM Router; 15-endpoint FastAPI backend; used to evaluate Leasel across 1,240 test cases; **RAG pipeline evaluation** with ChromaDB + Pinecone vector databases.

Leasel — Semantic Data Model & Context Engineering for AI Agents

Context Engineering · Semantic Layer · Knowledge Graph

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, hybrid RAG (TF-IDF + semantic reranking), SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Designed a **structured knowledge graph + semantic metadata layer** so 8 LangGraph AI agents accurately understand domain data (lease clauses, compliance rules, policy configurations) — same pattern as enabling LLMs to reliably query institutional data.
- **Context engineering**: section-aware chunking, citation metadata, structured entity resolution, and confidence-gated HITL escalation ensure agents reason over ground truth, not hallucinations; 100% citation correctness, 93.6% precision across 1,240-case eval harness. LLM reasons; deterministic logic decides every regulated call.

AI on FHIR — Natural Language to Structured Query (Text-to-API)

Text-to-SQL Pattern · HIPAA Governance

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- Built an NLP pipeline translating English/Spanish clinical questions into structured FHIR API calls via LLM + embeddings — the same **text-to-SQL / natural-language-to-query** pattern BU IS&T needs for institutional data; HIPAA RBAC guardrails; ~60% reduction in manual clinical lookup time.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Analytics Engineering & Data Architecture: dbt (semantic layer, semantic models, data modeling), Dagster (pipeline orchestration), Apache Iceberg (cloud lakehouse), Trino (query engine), SQL/PL-SQL (9+ yrs), Python (9+ yrs), Data Architecture, ETL Pipelines, Semantic Layer Design, Metrics Frameworks, Metadata Management, Query Optimization, Schema Design, Snowflake, PostgreSQL, MySQL, Oracle, SQLite, Redis, MongoDB

LLM Integration & Context Engineering: Text-to-SQL (LLM), text to SQL, RAG Pipelines, Context Engineering, MCP Integrations, LLM Accuracy Measurement, LLM-as-Judge, Semantic Search, Embeddings, Hybrid Retrieval, LangGraph StateGraph, LangChain, Multi-Agent Architectures, HITL Escalation, Guardrails, Prompt Engineering, Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, ChromaDB, Pinecone, Weaviate, Elasticsearch

Cloud, DevOps & Data Governance: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, HIPAA/PHI Compliance, HUD Regulatory Compliance, Data Privacy, Automated Testing, Code Review

Other Engineering: FastAPI, Flask, Django, Java, Spring Boot, TypeScript, JavaScript, Node.js, React, PHP, Bash, RabbitMQ, Kafka, Celery, scikit-learn, XGBoost, PyTorch, HuggingFace, BERT, spaCy, PySpark